End-Term Sample Question Paper - ANSWER KEY

Course Code: CSE563 Course Title: APPLIED MATHEMATICS

Maximum Marks: 60 Duration: 2 Hours Set A - all six units, theory and numerical

Academic Task Type: End-Term Sample Date: _____ Registration Number: _____

Instructions: answer all questions. Part A carries 2 marks each, Part B 5 marks each and Part C 10 marks each; in Part C the marks of each sub-part are shown in brackets. Show all working for numerical parts — an unsupported final answer earns no marks. The unit, course outcome and Bloom level are printed against every question.

For the instructor. Every numerical result in this key was recomputed and verified before printing; the marks in brackets give the part-wise marking scheme.

Part A - Very Short Answer Questions (5 x 2 = 10 Marks)

Q1. State the division algorithm, and state the exact condition under which an integer a has a multiplicative inverse modulo m. **(2 Marks)**

[Unit I · CO1 · L1 - Remember · Theory]

For any integer a and positive integer d there exist **unique** integers q and r with $a = dq + r$ and $0 \le r < d$. (1 mark)

$a^{-1} \bmod m$ exists **if and only if gcd(a, m) = 1**. (1 mark)

Q2. For $u = (1, -2, 2)$ and $v = (3, 0, 4)$, compute $u \cdot v$ and $\|u\|$. **(2 Marks)**

[Unit II · CO2 · L2 - Understand · Numerical]

$u \cdot v = 1(3) + (-2)(0) + 2(4) = 3 + 0 + 8 = \mathbf{11}$. (1 mark)

$\|u\| = \sqrt{(1 + 4 + 4)} = \sqrt{9} = \mathbf{3}$. (1 mark)

Q3. Two fair dice are rolled. Find the probability that the sum is 10, listing the favourable outcomes. **(2 Marks)**

[Unit III · CO3 · L2 - Understand · Numerical]

Favourable: (4,6), (5,5), (6,4) — **3** of the 36 outcomes. (1 mark)

$P(\text{sum} = 10) = 3/36 = \mathbf{1/12} \approx 0.0833$. (1 mark)

Q4. State the handshaking lemma, and use it to find the number of edges in a graph whose degree sequence is (4, 3, 3, 2, 2, 2). **(2 Marks)**

[Unit IV · CO4 · L1 - Remember · Theory]

Handshaking lemma: the degrees of all vertices sum to **twice** the number of edges, $\Sigma \deg(v) = 2E$. (1 mark)

$\Sigma \deg = 4+3+3+2+2+2 = 16 = 2E$, so $\mathbf{E = 8}$. (1 mark)

Q5. Define the notations O, Ω and Θ, and state what each one asserts about a running time. **(2 Marks)**

[Unit V · CO5 · L2 - Understand · Theory]

$f(n) = O(g(n))$ if $f(n) \le c \cdot g(n)$ beyond some n_0 — an **upper** bound. $f(n) = \Omega(g(n))$ if $f(n) \ge c \cdot g(n)$ — a **lower** bound. (1 mark)

$f(n) = \Theta(g(n))$ if both hold with different constants — a **tight** bound. (1 mark)

Part B - Short Answer Questions (4 x 5 = 20 Marks)

Q6. For the array [−3, 2, −1, 5, −4, 2], determine the maximum subarray sum and the subarray achieving it. Show the running best at each position, and state the running time of the scan. **(5 Marks)**

[Unit VI · CO6 · L3 - Apply · Numerical]

Kadane's scan keeps $cur = \max(x, cur + x)$ and $best = \max(best, cur)$. (1 mark)

Running best position by position: −3, 2, 2, **6**, 6, 6. (2 marks)

Maximum = **6**, from the subarray **[2, −1, 5]**. (1.5 marks)

The scan visits each element once, so it runs in **Θ(n)** — better than the Θ(n log n) divide-and-conquer solution. (0.5 marks)

Q7. Use the Euclidean algorithm to find gcd(252, 105), then use the extended Euclidean algorithm to find integers x and y with 252x + 105y = gcd(252, 105). Verify your answer. **(5 Marks)**

[Unit I · CO1 · L3 - Apply · Numerical]

252 = 2(105) + 42; 105 = 2(42) + 21; 42 = 2(21) + 0, so **gcd = 21**. (2 marks)

Back-substitute: 21 = 105 − 2(42) = 105 − 2(252 − 2(105)) = 5(105) − 2(252). (2 marks)

So **x = −2, y = 5**. Check: −2(252) + 5(105) = −504 + 525 = 21. Correct. (1 mark)

Q8. State Bayes' theorem, define prior, likelihood and posterior, and explain in two sentences why a highly accurate screening test can still produce mostly false alarms. **(5 Marks)**

[Unit III · CO3 · L2 - Understand · Theory]

P(D | +) = P(+ | D) P(D) / P(+), where P(+) is obtained by the law of total probability. (2 marks)

Prior P(D): belief before the evidence. **Likelihood** P(+ | D): probability of the observed result if the hypothesis holds. **Posterior** P(D | +): revised belief after the evidence. (2 marks)

When the condition is rare, the large healthy group generates more false positives than the small ill group generates true positives, so most positives are false — the **base-rate fallacy**. Sensitivity alone says nothing; the prior dominates. (1 mark)

Q9. Compare linear search and binary search on a sorted array of n elements: give the worst-case comparison count for each, state the precondition binary search needs, and state the lower bound for comparison-based sorting with a one-line reason. **(5 Marks)**

[Unit V · CO5 · L4 - Analyse · Theory]

Linear search: **n** comparisons in the worst case; binary search: **$\lfloor \log_2 n \rfloor + 1$**. (2 marks)

Binary search requires the array to be **sorted**; it is useless otherwise. (1 mark)

Any comparison-based sort needs **$\Omega(n \log n)$** comparisons: its decision tree must have n! leaves, and a binary tree with n! leaves has height at least $\log_2(n!) = \Omega(n \log n)$. (2 marks)

Part C - Long Answer Questions (3 x 10 = 30 Marks)

Q10. Principal Component Analysis. **(10 Marks)**

[Unit II · CO2 · L4 - Analyse · Theory + Numerical]

(a) State the main steps of PCA and explain why the data must be centred before the covariance matrix is formed. *[5]*

(b) For the points (2,1), (3,2), (4,4), (5,3), (6,5): compute the mean, the 2×2 covariance matrix (divide by n − 1), both eigenvalues, and the percentage of variance carried by PC1. *[5]*

(a) Centre the data; form the covariance matrix; find its eigenvalues and eigenvectors; order the eigenvectors by decreasing eigenvalue; project onto the leading ones. (3 marks)

PCA seeks directions of greatest **variance**, and variance is measured about the mean. Without centring the first component merely points at the mean vector rather than at the direction of spread. (2 marks)

(b) Mean = (4, 3). Centred: (−2,−2), (−1,−1), (0,1), (1,0), (2,2). (1 mark)

$\Sigma\tilde{x}^2 = 10$, $\Sigma\tilde{y}^2 = 10$, $\Sigma\tilde{x}\tilde{y} = 9$, so $S = \frac{1}{4}[[10, 9], [9, 10]] =$ **[[2.5, 2.25], [2.25, 2.5]]**. (2 marks)

$\lambda^2 - 5\lambda + (6.25 - 5.0625) = 0$ gives $\lambda = (5 \pm 4.5)/2$, so **$\lambda_1 = 4.75$** and **$\lambda_2 = 0.25$** (sum 5 = trace, check). (1.5 marks)

PC1 carries 4.75/5 = **95 per cent** of the variance. (0.5 marks)

Q11. Shortest paths in a weighted directed graph. **(10 Marks)**

[Unit IV · CO4 · L4 - Analyse · Theory + Numerical]

(a) Describe the central idea of Dijkstra's algorithm, and explain why it can return a wrong answer when an edge weight is negative. *[5]*

(b) Apply it from A to the graph with edges A→B = 6, A→C = 2, C→B = 3, B→D = 1, C→D = 7, D→E = 4. Give the distance to every vertex, the finalisation order and each predecessor. *[5]*

(a) Repeatedly take the unvisited vertex of smallest tentative distance, fix that distance as final, and relax its outgoing edges. (3 marks)

Correctness assumes no later path can be cheaper — i.e. that extending a path never reduces its cost. A negative edge breaks that assumption, so a vertex may be finalised before a cheaper route through the negative edge is discovered. Bellman–Ford is used instead. (2 marks)

(b) d[A] = 0, d[C] = **2**, d[B] = **5** via A–C–B (2+3 beats the direct edge 6), d[D] = **6** via B (5+1 beats 2+7 = 9), d[E] = **10**. (3 marks)

Finalisation order: **A(0), C(2), B(5), D(6), E(10)**. (1 mark)

Predecessors: π[C] = A, π[B] = C, π[D] = B, π[E] = D. (1 mark)

Q12. Greedy and dynamic programming on the knapsack problem. **(10 Marks)**

[Unit VI · CO6 · L5 - Evaluate · Theory + Numerical]

(a) State the greedy-choice property and optimal substructure, and say which of the two knapsack versions has each. *[5]*

(b) Items are (2 kg, ₹20), (4 kg, ₹32) and (6 kg, ₹36) with capacity 10 kg. Solve the fractional version by the greedy ratio rule, give the true 0/1 optimum, and comment on the gap. *[5]*

(a) Greedy-choice property: a globally optimal solution can be reached by taking the locally best option at each step. **Optimal substructure:** an optimal solution contains optimal solutions to its subproblems. (3 marks)

The **fractional** knapsack has both, so greedy is optimal. The **0/1** version has optimal substructure but **not** the greedy-choice property, so it needs dynamic programming. (2 marks)

(b) Ratios: 20/2 = **10**, 32/4 = **8**, 36/6 = **6**. (1 mark)

Greedy fractional: all of item 1 (2 kg, 20), all of item 2 (4 kg, 32), then 4 kg of item 3 = (4/6)(36) = 24. Total = **₹76**. (2 marks)

0/1 optimum: items 2 and 3 weigh 4 + 6 = 10 kg exactly and give 32 + 36 = **₹68**. (1.5 marks)

The 0/1 answer is **8 lower**, and it does **not** contain the best-ratio item at all — committing 2 kg to item 1 would block the pair that fills the knapsack exactly. That is precisely why greedy fails once items are indivisible. (0.5 marks)

----- End of Answer Key -----